COMPETENCY STANDARD
FOR
Advanced Pipe Fitting for Gas and Onshore Rigs

(ENERGY SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: https://sicip.gov.bd/

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.
Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-BPI-APF-01-G	Apply Occupational Health and Safety (OHS) Practice in the Workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-BPI-APF-02-G	Carry Out Workplace Interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	10
SICIP-BPI-APF-03-G	Operate In a Team Environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	10
		Total Hour	**30 Hrs.**

Sector Specific Competencies (10 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-BPI-APF-01-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	10
		Total Hour	**10 Hrs.**

Occupation Specific Competencies (320 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-BPI-APF-01-O	Apply Safety and Environmental Rules in Oil and Gas Work	1. Identify hazards in gas and rig operations. 2. Use PPE and safety equipment. 3. Follow hot-work, lifting, and confined-space procedures. 4. Identify environmental and emergency rules. 5. Maintain a clean and safe worksite.	10
SICIP-BPI-APF-02-O	Interpret Drawing, Specification and Pipe Layout Designs	1. Interpret symbols, terminology, and dimensions. 2. Interpret P&ID, isometric, and blueprint drawings. 3. Describe drawing specifications and requirements. 4. Explain layout plans stress, vibration, and expansion. 5. Identify alignment and geometric accuracy in drawings.	40
SICIP-BPI-APF-03-O	Identify Pipe Materials and Fittings for Oil & Gas Work	1. Interpret ASME, ASTM, ANSI and API Codes and standards/ specifications. 2. Identify carbon steel, stainless steel, and alloy steel pipes. 3. Recognize pipe fittings. 4. Identify materials based on pressure, temperature, and corrosion conditions.	25
SICIP-BPI-APF-04-O	Prepare Tools, Pipes, logistics and Worksite for Installation	1. Select tools, equipment, and pipe materials. 2. Measure, mark and Fabricate pipes to required dimensions. 3. Bevel, grind, and prepare pipe ends. 4. Set up worksite for safe	50

Code	Unit of Competency	Elements of Competency	Duration (hours)
		operations. 5. Inspect pipes and remove defects.	
SICIP-BPI-APF-05-O	Perform Pipe Fittings, Joining and Alignment	1. Identify SMAW methods. 2. Position pipes as per drawings. 3. Fabricate, bend, and thread pipes as required. 4. Align joints with clamps, jigs, or supports. 5. Set root gap, face, and alignment tolerances. 6. Perform SMAW Welding.	120
SICIP-BPI-APF-06-O	Install and Support Pipeline Systems	1. Install pipes, flanges, valves, and fittings. 2. Assemble gaskets, bolts, and joint parts. 3. Apply torque and tightening steps. 4. Install supports, hangers, and guides. 5. Verify alignment and required slope.	50
SICIP-BPI-APF-07-O	Understand inspection and testing Procedures	1. Interpret inspection 2. Perform testing of weld 3. Clean and store equipment	25
		Total Hours	**0 Hrs.**

COMPETENCY STANDARD: Advanced Pipe Fitting for Gas and Onshore Rigs

Generic Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-BPI-APF-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **OHS policies** and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are interpreted in the workplace including **personal protective equipment (PPE).** 1.5 Common health issues are recognized. 1.6 Common safety issues are identified and applied.
3. Report hazards and risks	1.1 Hazards and risks are identified. 1.2 Hazards and risks assessment and controls are interpreted 1.3 Hazards and risk assessment are reported
4. Respond to emergencies	1.1 Respond to alarms and warning devices. 1.2 **Emergency response plans and procedures** are responded to. 1.3 **First aid procedures** during emergency situations are identified.

Range of variables:

Variables	**Range** (may include but not limited to)
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective	2.1 Safety glasses

equipment	2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	1.8 Identifying OHS policies and procedures 1.9 Applying personal health and safety practices 1.10 Reporting hazards and risks 1.11 Responding to emergencies
3. Underpinning Attitudes	1.12 Commitment to occupational health and safety 1.13 Promptness in carrying out activities 1.14 Sincere and honest to duties 1.15 Environmental concerns 1.16 Eagerness to learn 1.17 Tidiness and timeliness 1.18 Respect for rights of peers and seniors in the workplace 1.19 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.20 Workplace (simulated or actual) 1.21 Workplace documents, signs and symbols 1.22 Codes of conduct 1.23 Projector 1.24 Learning manual

	1.25 Tools, equipment and facilities appropriate to the process or activities. 1.26 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 10 hrs.	Unit Code: SICIP-BPI-APF-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **<u>courteous manner</u>** to gather and convey information. 1.4 **<u>Workplace procedures and matters</u>** are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from **<u>appropriate sources</u>**. 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening

	1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	1.1 Notes 1.2 Arranging a meeting 1.3 Agenda 1.4 Simple reports such as progress and incident reports 1.5 Job sheets 1.6 Operational manuals 1.7 Brochures and promotional material 1.8 Visual and graphic materials 1.9 Standards 1.10 OHS information 1.11 Signs
3. Appropriate sources	1.1 Human Resources (HR) Department 1.2 Managers 1.3 Supervisors 1.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	1.5 Demonstrating workplace communication and etiquette 1.6 Interpreting workplace instructions and symbols 1.7 Demonstrating active participation in workplace meeting 1.8 Applying professional ethics at work
3. Underpinning Attitudes	1.9 Commitment to occupational health and safety 1.10 Promptness in carrying out activities 1.11 Sincere and honest to duties 1.12 Environmental concerns 1.13 Eagerness to learn 1.14 Tidiness and timeliness 1.15 Respect for rights of peers and seniors in the workplace 1.16 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-BPI-APF-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through **<u>sharing information</u>** or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	1.1. Identifying own role and responsibilities within team 1.2. Communicating and co-operating with team members 1.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	1.4. Commitment to occupational health and safety 1.5. Promptness in carrying out activities 1.6. Sincere and honest to duties 1.7. Eagerness to learn 1.8. Tidiness and timeliness 1.9. Environmental concerns 1.10. Respect for rights of peers and seniors at workplace 1.11. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 1.12. Workplace (simulated or actual) 1.13. Workplace documents, Projector 1.14. Learning manual 1.15. Tools, equipment and facilities appropriate to the process or activities. 1.16. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration

	2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **WORK WITH HAND AND POWER TOOLS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-BPI-APF-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1. Appropriate **<u>hand tools</u>** are selected. 2.2. Safety precautions are ensured before using hand tools. 2.3. Unsafe or faulty hand tools are identified and marked for repair. 2.4. **<u>Measuring tools</u>** are checked and calibrated before use. 2.5. Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate **<u>power tools</u>** are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported.

	4.3. Appropriate lubricant is applied after use and prior to storage. 4.4. Defective hand and power tools are inspected and repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.

Range of Variables

Variable	**Range** May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification

	1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: APPLY SAFETY AND ENVIRONMENTAL RULES IN OIL AND GAS WORK	Nominal Duration: 10 Hrs.	Unit Code: SICIP-BPI-APF-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply safety and environmental rules in oil and gas work. It specifically includes the task of identifying hazards in gas and rig operations, using PPE and safety equipment, following hot-work, lifting, and confined-space procedures, identifying environmental and emergency rules and maintaining a clean and safe worksite.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify hazards in gas and rig operations.	1.1 Gas leaks and **<u>fire hazards</u>** are recognized. 1.2 **<u>Mechanical and equipment-related hazards</u>** are identified. 1.3 Unsafe behaviors and procedural violations are interpreted. 1.4 Worksite conditions are assessed to identify potential hazards.
2. Use PPE and safety equipment.	2.1 **<u>Required PPE</u>** is selected according to job and hazard assessments. 2.2 PPE is inspected for proper condition and functionality. 2.3 **<u>Safety equipment</u>** is worn and adjusted correctly. 2.4 Defective or damaged PPE is removed from use and reported. 2.5 PPE is used consistently throughout all work activities.
3. Follow hot-work, lifting, and confined-space procedures.	3.1 Hot-work permits and requirements are reviewed and followed. 3.2 Lifting plans and load limits are checked before operations. 3.3 Confined-space entry procedures and monitoring steps are applied. 3.4 Safety controls and equipment are used according to procedures.
4. Identify environmental and emergency rules.	4.1 Environmental regulations for gas and rig operations are reviewed. 4.2 Spill, waste, and pollution risks are identified. 4.3 Emergency response procedures are recognized and understood. 4.4 **<u>Emergency equipment</u>** locations and alarms are identified.
5. Maintain a clean and safe worksite.	5.1 Tools and materials are kept organized to prevent

	hazards. 5.2 Waste and debris are removed according to site procedures. 5.3 Slippery, obstructed, or unsafe areas are corrected or reported. 5.4 Worksite cleanliness is maintained throughout operations.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Fire hazards	1.1 Gas leaks 1.2 Sparks from welding or cutting 1.3 Static electricity 1.4 Faulty electrical tools 1.5 Flammable materials (rags, solvents) 1.6 Open flames or smoking 1.7 Overheated machinery 1.8 Leaking gas cylinders
2. Mechanical and equipment-related hazards	2.1 Moving machinery parts 2.2 Pinch points during pipe handling 2.3 Heavy pipe lifting and dropping 2.4 Equipment failure or malfunction 2.5 Pressure release from valves or fittings 2.6 Falling tools or materials 2.7 Slips on oily or uneven surfaces 2.8 Grinder or cutting wheel breakage 2.9 Vibration from power tools
3. Required PPE	3.1 Safety helmet 3.2 Safety goggles or face shield 3.3 Flame-resistant coveralls/ Apron 3.4 Safety gloves 3.5 Steel-toe safety boots 3.6 Hearing protection 3.7 Respiratory protection (mask or respirator) 3.8 High-visibility vest 3.9 Welding shield and welding gloves (for hot work)
4. Safety equipment	4.1 Fire extinguishers 4.2 Gas detectors 4.3 Fire and gas alarm system 4.4 Safety showers and eyewash stations 4.5 First aid kit 4.6 Breathing apparatus (SCBA) 4.7 Fire blankets
5. Emergency equipment	5.1 Emergency shutdown system (ESD) 5.2 Emergency eyewash station 5.3 Safety shower

	5.4 Stretcher or rescue stretcher 5.5 Fire extinguishers 5.6 Fire hose and hydrant system 5.7 Emergency alarm system

Curricular Content Guide

1. Underpinning Knowledge	1.1. Different types of hazards commonly found in gas and rig operations 1.2. Safety regulations and procedures 1.3. Various types of Personal Protective Equipment (PPE) 1.4. Regulations governing hot-work, lifting, and confined-space procedures operations in the gas and rig industry. 1.5. Environmental regulations and emergency procedures concerning gas and rig operations 1.6. Safety regulations for maintaining a clean and orderly worksite.
2. Underpinning Skills	2.1. Observing and assess the work environment for potential hazards 2.2. Selecting appropriate PPE based on the type of work being done 2.3. Performing hot-work tasks 2.4. Operating lifting equipment 2.5. Recognizing potential environmental hazards 2.6. Executing emergency procedures 2.7. Performing regular housekeeping tasks
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified hazards in gas and rig operations,

	1.2 used PPE and safety equipment, 1.3 followed hot-work, lifting, and confined-space procedures, 1.4 identified environmental and emergency rules 1.5 maintained a clean and safe worksite.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INTERPRET DRAWING, SPECIFICATION AND PIPE LAYOUT DESIGNS**	**Nominal Duration:** 40 Hrs.	**Unit Code:** SICIP-BPI-APF-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret drawing, specification and pipe layout designs. It specifically includes the task of interpreting symbols, terminology, and dimensions, interpreting P&ID, isometric, and blueprint drawings, describing drawing specifications and requirements, explaining layout plans stress, vibration, and expansion and identifying alignment and geometric accuracy in drawings.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret symbols, terminology, and dimensions.	1.1 Symbols, terminologies and dimensions are comprehended 1.2 The meaning of symbols and terms is accurately explained 1.3 Measurement knowledge is acquired and **unit conversions** process are explained. 1.4 Dimensions relevant to the job are identified and clarified.
2. Interpret P&ID, isometric, and blueprint drawings.	2. 1 The Piping & Instrumentation Diagram (P&ID), isometric, and blueprint drawings are interpreted according to process segment. 2. 2 The symbols, lines, and annotations used in the drawings are identified and understood. 2. 3 The layout, dimensions, and scale of the drawings are analyzed accurately. 2. 4 The drawings are used to identify key information required for pipe fitting activities. 2. 5 All relevant technical data from the drawings are extracted and applied appropriately to the task at hand. 2. 6 Drawing specifications, **terminologies** and requirements are explained.
3. Describe drawing specifications and requirements.	3. 1 Drawing specifications are reviewed to ensure all required details. 3. 2 Standard symbols and notations are identified and their meanings are understood to interpretation of the drawings. 3. 3 Drawing requirements are compared with project guidelines and regulatory standards.
4. Explain layout plans stress, vibration, and	4. 1 Stress factors are identified in the layout plan, considering the potential forces. 4. 2 Stress distribution is analyzed within the piping or

expansion.	structural system. 4. 3 Vibration effects are described within the layout plan. 4. 4 Expansion considerations are outlined in the layout plan. 4. 5 Mitigation methods are discussed for handling stress, vibration, and expansion issues.
5. Identify alignment and geometric accuracy in drawings.	5. 1 The alignment of components, lines, and structures in the drawings is identified. 5. 2 The geometric accuracy of shapes, angles, and dimensions in the drawings is assessed according to the specified standards. 5. 3 The appropriate tools and methods for verifying alignment and geometric accuracy are selected and applied. 5. 4 The correct interpretation of tolerances, fits, and clearances related to alignment and geometry is demonstrated. 5. 5 Corrective actions or recommendations for improving alignment and geometric accuracy are identified and communicated clearly.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Unit conversions	1.1 Inch to millimeter, Inch to centimeter, 1.2 Feet to meter, meter to feet 1.3 Feet to inch
2. Terminologies	2.1 Pipes 2.2 Valve (Ball, Plug, Glove, PCV, FCV) 2.3 Fittings (Elbow, Tee, Reducer, Union, Pipe Cap, Cross) 2.4 Hammer/WECO joint, Swivel joint, 2.5 Instruments 2.6 Symbol

Curricular Content Guide

1. Underpinning Knowledge	1.1. Understand standard symbols, terminology and dimensions used in technical drawings. 1.2. Function and structure of different types of Piping and Instrumentation Diagrams (P&ID), isometric drawings, and blueprints. 1.3. Understand the specification of drawings. 1.4. Describe stress, vibration, and expansion. 1.5. Importance of geometric accuracy in technical drawings. 1.6. Understand the standards and guidelines for ensuring alignment in drawings.
2. Underpinning Skills	2.1. Recognizing and interpreting common symbols used in

	technical drawings 2.2. Applying technical terminology 2.3. Interpreting the layout and dimensions of a drawing 2.4. Interpreting P&ID diagrams and isometric drawings 2.5. Using dimensioning techniques to ensure correct measurements. 2.6. Identifying standard symbols and notations 2.7. Describing mitigation methods for handling stress, vibration, and expansion issues. 2.8. Identifying and resolving alignment issues 2.9. Checking for geometric accuracy in technical drawings,
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted symbols, terminology, and dimensions 1.2 interpreted P&ID, isometric, and blueprint drawings 1.3 described drawing specifications and requirements. 1.4 explained layout plans stress, vibration, and expansion. 1.5 identified alignment and geometric accuracy in drawings.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IDENTIFY PIPE MATERIALS AND FITTINGS FOR OIL & GAS WORK**	Nominal Duration: 25 Hrs.	Unit Code: SICIP-BPI-APF -03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify pipe materials and fittings for oil & gas work. It specifically includes the task of interpreting ASME, ASTM, ANSI and API Codes and standards/ specifications, identifying carbon steel, stainless steel, and alloy steel pipes, recognizing pipe fittings and identifying materials based on pressure, temperature, and corrosion conditions.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret ASME, ASTM, and API Codes and standards/ specifications.	1.1 **<u>Relevant ASME, ASTM, and API codes/standards</u>** are identified from project requirements. 1.2 Code sections and terminology are reviewed and interpreted accurately. 1.3 Material, fabrication, and testing requirements are understood from the standards. 1.4 Acceptance criteria and limitations are interpreted according to code specifications.
2. Identify carbon steel, stainless steel, and alloy steel pipes.	2.1 **<u>Pipe materials</u>** are identified through markings, color codes, or labels. 2.2 Visual characteristics of carbon steel, stainless steel, and alloy steel are recognized. 2.3 Material grade and composition are interpreted from documentation. 2.4 Surface condition and corrosion resistance are identified. 2.5 Use of carbon steel, stainless steel, and alloy steel are identified as per process conditions.
3. Recognize pipe fittings.	3.1 **<u>Common pipe fittings</u>** are identified by shape and function. 3.2 Flange types and pressure ratings are identified from markings. 3.3 Threaded, welded, and socket-weld fittings are distinguished. 3.4 Material type and suitability of fittings are checked against specifications.
4. Identify materials based on pressure, temperature, and corrosion conditions.	4.1 **<u>Operating pressure and temperature</u>** requirements are reviewed from project data. 4.2 Material properties are compared with service conditions.

	4.3 Corrosion risks and environmental factors are identified. 4.4 Suitable materials are selected according to code requirements.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Relevant ASME, ASTM, and API codes/standards	**American Society for Mechanical Engineers (ASME)** 1.1 ASME B31.3 – Process Piping 1.2 ASME B31.8 – Gas Transmission and Distribution 1.3 ASME Section IX – Welding Qualifications 1.4 ASME B16.25 – Buttwelding Ends 1.5 ASME B16.9 – Factory-Made Wrought Fittings **American Society for Testing Materials (ASTM)** 1.6 ASTM A53 – Steel Pipe 1.7 ASTM A106 – Seamless Carbon Steel Pipe 1.8 ASTM A312 – Stainless Steel Pipe 1.9 ASTM A234 – Carbon & Alloy Steel Fittings 1.10 ASTM A105 – Carbon Steel Forgings 1.11 ASTM A193 – Alloy Steel Bolting 1.12 ASTM A194 – Carbon & Alloy Steel Nuts **American Petroleum Institute (API)** 1.13 API 5L – Line Pipe 1.14 API 1104 – Pipeline Welding 1.15 API 570 – Piping Inspection 1.16 API 6D – Pipeline Valves
2. Pipe materials	2.1 Carbon steel 2.2 Stainless steel 2.3 Alloy steel
3. Common pipe fittings	3.1 Elbow 3.2 Tee 3.3 Reducer 3.4 Coupling 3.5 Union 3.6 Flange 3.7 Cap 3.8 Plug and socket 3.9 Valve 3.10 Nipple 3.11 Saddle/weld-o-let
4. Operating pressure and temperature	4.1 Maximum operating pressure 4.2 Minimum operating pressure 4.3 Maximum operating temperature 4.4 Minimum operating temperature 4.5 Pressure rating of pipes and fittings 4.6 Temperature rating of materials

Curricular Content Guide

1. Underpinning Knowledge	1.1. Understand the structure and content of ASME, ASTM, ANSI, and API codes. 1.2. The specific roles of these codes play in defining safety, quality, and performance standards 1.3. Characteristics and Applications of carbon steel, stainless steel, and alloy steel pipes. 1.4. Differences of composition, mechanical properties, and applications of steel pipes 1.5. Different types of fittings used in plumbing and industrial piping systems. 1.6. Understand different materials perform under varying temperature and pressure conditions.
2. Underpinning Skills	2.1. Applying ASME, ASTM, ANSI, and API codes. 2.2. Understanding the technical language and requirements of these standards 2.3. Determining the appropriate pipe material for specific industrial applications. 2.4. Identifying different types of pipe fittings and their specific applications in piping systems. 2.5. Ability to evaluate pressure and temperature specifications and select materials that will perform well under these conditions. 2.6. Assessing environmental conditions to select appropriate materials.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted ASME, ASTM, ANSI and API Codes and standards/ specifications 1.2 identified carbon steel, stainless steel, and alloy steel

	pipes 1.3 recognized pipe fittings 1.4 identified materials based on pressure, temperature, and corrosion conditions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PREPARE PIPES, TOOLS, LOGISTICS AND WORKSITE FOR INSTALLATION**	**Nominal Duration:** 50 Hrs**.**	**Unit Code:** SICIP-BPI-APF-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to prepare tools, pipes, logistics and worksite for installation. It specifically includes the tasks of selecting tools, equipment, and pipe materials, measuring, marking and fabricating pipes to required dimensions, beveling, grinding, and preparing pipe ends, setting up worksite for safe operations and inspecting pipes and remove defects.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Select tools, equipment, logistics and pipe materials.	1.1 The **appropriate tools**, equipment, **logistics** and pipe materials are selected based on the requirements of the task. 1.2 The tools and equipment are chosen based on their suitability for the task, considering factors such as functionality, safety, and efficiency. 1.3 The pipe materials are selected based on their resistance to corrosion, temperature, pressure, and other environmental factors. 1.4 The correct sizes, types, and quantities of tools, equipment, and pipe materials are identified and confirmed for the task. 1.5 The relevant industry standards, codes, and regulations are followed in the selection process to ensure compliance and quality.
2. Measure, mark and cut pipes to required dimensions.	2. 1 The correct tools and equipment for measuring, marking, and cutting pipes are selected based on the type of pipe and the required task. 2. 2 The dimensions for cutting the pipes are accurately measured and marked according to the specified requirements. 2. 3 The marking is done clearly and precisely to ensure correct cutting and alignment. 2. 4 The pipes are cut using the appropriate technique, ensuring a clean and accurate cut without damage to the material. 2. 5 The cuts are checked to verify that they meet the required dimensions and tolerances. 2. 6 Safety procedures and precautions are followed during the measuring, marking, and cutting process to prevent accidents or injury. 2. 7 The work area is kept clean and organized throughout the process, with tools and materials stored safely after

	use.
3. Bevel, grind, and prepare pipe ends.	3.1 The appropriate tools and equipment for beveling, grinding, and preparing pipe ends are selected. 3.2 The pipe ends are carefully beveled to the required angle, ensuring a clean, uniform bevel suitable for welding or joining. 3.3 The pipe ends are ground to remove any rough edges, burrs, or imperfections, ensuring smooth surfaces for proper sealing or fitting. 3.4 The correct grinding technique is applied to avoid over-heating or damaging the pipe material. 3.5 The dimensions and angles of the bevels are verified to ensure they meet the specified requirements. 3.6 The pipe ends are inspected for cleanliness and free from contaminants, such as oil, rust, or dirt, before proceeding next phase of the piping system assembly or installation. 3.7 Safety protocols are followed throughout the beveling, grinding, and preparation process to minimize risks of injury or material damage.
4. Set up worksite for safe operations.	4.1 The worksite is assessed for potential hazards and risks, and appropriate safety measures are identified. 4.2 The work area is organized and clear of unnecessary obstructions to ensure a safe and efficient workspace. 4.3 Safety signage, barriers, or warning systems are put in place to alert workers to potential hazards in the worksite. 4.4 Proper ventilation or other environmental controls are set up if required, especially in confined spaces or areas with hazardous fumes. 4.5 All relevant safety procedures, regulations, and industry standards are reviewed and followed to ensure compliance. 4.6 Emergency procedures and exits are clearly marked, and the necessary first aid equipment is available and accessible. 4.7 Communication systems are established to ensure clear and effective coordination among workers on-site.
5. Inspect pipes and remove defects.	5.1 The pipes are thoroughly inspected for defects, including porous, cracks, misalignment, or other physical damage. 5.2 **Defective areas** are identified and marked for repair or removal, following established procedures. 5.3 The appropriate tools and equipment are selected and used to remove defects, such as grinding, polishing, or cutting tools. 5.4 Defects are removed or repaired to restore the pipe's functionality and ensure it meets the required standards for use.

	5.5 The pipe surface is checked for smoothness and free of sharp edges or burrs that could cause further issues.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. **<u>appropriate tools</u>**	1.1 Grinder 1.2 Cutting Machine (Power saw, Cutting Disc) 1.3 Gas Cutting 1.4 Measuring tap 1.5 Welding Generator 1.6 Welding equipment 1.7 Scriber 1.8 Wire Brus 1.9 Center Punch 1.10 Hammer
2. Measuring tools	2.1 Vernier Caliper 2.2 Steel rule 2.3 Thickness gauge 2.4 Micro Meter 2.5 Level gauge 2.6 Laser tools 2.7 High precision level gauge
3. Logistics	3.1 Overhead crane 3.2 Engin crane 3.3 Forklift 3.4 Chain tong 3.5 Brackets
4. Defective areas	4.1 Welding joint 4.2 Flange joint 4.3 Misalignment 4.4 Over tightness

Curricular Content Guide

1. Underpinning Knowledge	1.1. Different types of materials used in pipe construction and their properties 1.2. Various tools and equipment used in pipe fabrication 1.3. Measurement techniques use in pipe fabrication 1.4. Understand different fabrication methods 1.5. Purpose and techniques for beveling and grinding pipe ends 1.6. The role of pipe end preparation in preventing weld defects 1.7. Understand the safe worksite practices. 1.8. Steps involved in assessing risks at the worksite and implementing control measures

	1.9. Different inspection methods used to detect defects in pipes 1.10. Techniques for removing defects from pipes
2. Underpinning Skills	2.1. Selecting the appropriate pipe material based on the specific needs of the project 2.2. Operating tools and equipment for pipe fabrication 2.3. Performing routine maintenance 2.4. Measuring and marking pipes based on project specifications 2.5. Fabricating pipes to required dimensions 2.6. Beveling and grinding pipe ends to ensure proper fitment for welding. 2.7. Performing pre-welding inspections to confirm proper preparation and alignment of pipe ends 2.8. Setting up a worksite for safe operations. 2.9. Performing a risk assessment and implementing safety procedures on the worksite. 2.10. Inspecting pipes for defects using various testing methods. 2.11. Removing defects from pipes through appropriate repair techniques.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 selected tools, equipment, and pipe materials 1.2 measured, mark and fabricat pipes to required dimensions 1.3 beveled, grinded, and prepared pipe ends 1.4 set up worksite for safe operations 1.5 inspected pipes and remove defects.

2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM PIPE FITTINGS, JOINING AND ALIGNMENT**	**Nominal Duration:** 120 Hrs.	**Unit Code:** SICIP-BPI-APF-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform pipe fittings, joining and alignment. It specifically includes the tasks of identifying SMAW methods, positioning pipes as per drawings, fabricating, bending, and threading pipes as required, aligning joints with clamps, jigs, or supports, setting root gap, face, and alignment tolerances and performing SMAW welding.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify SMAW methods.	1.1 The Shielded Metal Arc Welding (SMAW) methods are identified. 1.2 The key characteristics, advantages, and limitations of both SMAW welding processes are explained. 1.3 The equipment, tools, and materials required for SMAW welding are understood. 1.4 The differences in the heat sources, filler materials, and shielding gases used in SMAW welding are identified. 1.5 The necessary safety precautions specific to SMAW welding are understood.
2. Position pipes as per drawings.	2.1 The pipes are positioned according to the specified dimensions and orientations shown in the drawings. 2.2 The correct tools, equipment, and supports are used to position the pipes securely and safely, ensuring alignment with the design specifications. 2.3 Any obstacles or constraints that may affect pipe positioning, as indicated in the drawings, are identified and managed effectively. 2.4 The pipe supports, brackets, and anchors are installed as per the drawings, ensuring proper stability and support during operation.
3. Fabricate, bend, and thread pipes as required.	3.1 The correct tools and equipment are selected and inspected for cutting, bending, and threading pipes. 3.2 The pipes are accurately measured and marked to the required dimensions before cutting, bending, or threading. 3.3 The bending process is carried out according to the required angle or radius, ensuring that the pipe does not become distorted or weakened. 3.4 The pipes are threaded using the correct threading

	tools and techniques, ensuring that the threads are clean, even, and free from defects. 3.5 The **bend** radius and **thread** pitch are checked to ensure they meet the specified requirements and are compatible with fittings and connectors.
4. Align joints with clamps, jigs, or supports.	4.1 The correct clamps, jigs, or supports are selected based on the type and size of the pipe. 4.2 The clamps, jigs, or supports are securely set up to hold the joint in place, preventing movement during welding or assembly. 4.3 The joint is adjusted as necessary to ensure that the alignment meets the specified standards and tolerances. 4.4 Any discrepancies in the alignment are identified and corrected before proceeding with further welding or assembly.
5. Set root gap, face, and alignment tolerances.	5.1 The root gap, face, and alignment tolerances are determined based on the specifications provided in the drawings. 5.2 The root gap is set accurately, ensuring the correct spacing between the pipe edges for proper welding penetration and strength. 5.3 The **pipe** faces are aligned correctly to ensure even surfaces and proper fit during welding or joining. 5.4 The tools and equipment used to measure and set root gap, face, and alignment tolerances are correctly selected and calibrated. 5.5 The **tolerances** are verified using appropriate measuring tools, such as gauges or calipers, ensuring they meet the specified standards.
6. Perform SMAW Welding.	6.1 The appropriate welding equipment, electrodes, and materials for both SMAW (Shielded Metal Arc Welding) are selected. 6.2 The welding procedures for SMAW are followed according to the relevant standards and specifications, ensuring proper technique and safety. 6.3 The welding parameters (e.g., current, voltage, speed) for SMAW are correctly set according to the material type, thickness, and welding position. 6.4 Proper SMAW are used to protect the weld from contamination during the process.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Bend	1.1 Long bend 1.2 Short bend

2. Thread	2.1 NPT 2.2 BST 2.3 BTC (API Buttress thread tap) 2.4 BSPP (British standard parallel thread)
3. Pipe	3.1 Seamless 3.2 Welded
4. Tolerances	4.1 ASME B-31.3 4.2 PFI ES-3

Curricular Content Guide

1. Underpinning Knowledge	1.1. Understand the principles of SMAW 1.2. Advantages and limitations of SMAW compared to other welding methods. 1.3. Interpret technical drawings 1.4. Various techniques for positioning pipes during fabrication 1.5. Different pipe fabrication methods 1.6. Tools and techniques for bending and threading pipes. 1.7. Various tools and equipment used to align pipe joints. 1.8. Techniques for achieving correct joint alignment. 1.9. Tolerances in Pipe Welding 1.10. Methods for measuring and setting root gap, face, and alignment tolerances 1.11. Process of setting up the welding machine, preparing the work area, and selecting the correct electrode. 1.12. Understand the key indicators of good SMAW weld. 1.13. Methods for inspecting and testing the welds.
2. Underpinning Skills	2.1 Selecting the appropriate SMAW method based on material type, thickness. 2.2 Applying SMAW techniques in different welding positions. 2.3 Positioning pipes accurately according to engineering drawings 2.4 Interpreting technical drawings and pipe layouts. 2.5 Fabricating, cutting, and bending pipes to the required dimensions. 2.6 Threading pipes using appropriate tools and techniques. 2.7 Aligning pipe joints with the use of clamps, jigs, or supports. 2.8 Adjusting pipe ends for proper fitment and ensure alignment. 2.9 Performing SMAW welds on various pipe materials and joint. 2.10 Inspecting welds for defects and making adjustments.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified SMAW methods 1.2 positioned pipes as per drawings 1.3 fabricated, bended, and threaded pipes as required 1.4 aligned joints with clamps, jigs, or supports 1.5 set root gap, face, and alignment tolerances 1.6 performed SMAW welding.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INSTALL AND SUPPORT PIPELINE SYSTEMS**	**Nominal Duration:** 50 Hrs.	**Unit Code:** SICIP-BPI-APF-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install and support pipeline systems. It specifically includes the tasks of installing pipes, flanges, valves, and fittings, assembling gaskets, bolts, and joint parts, applying torque and tightening steps, installing supports, hangers, and guides and verifying alignment and required slope.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install pipes, flanges, valves, and fittings.	1.1 The correct types, sizes, and quantities of **pipes, flanges, valves,** and fittings are selected asper specifications. 1.2 The installation process follows the engineering drawings, and relevant industry standards. 1.3 The pipes are positioned and supported securely, ensuring proper alignment, slope with flanges, valves, and fittings. 1.4 The flanges are aligned properly with the pipe ends, ensuring a tight, leak-free connection when bolted. 1.5 The valves are installed in the correct orientation and location, ensuring proper flow control within the system. 1.6 Fittings, such as elbows, tees, or couplings, are installed according to the design, ensuring that the pipe layout meets the required flow path. 1.7 The system is inspected for any potential defects or misalignments before finalizing the installation to ensure according to design and drawing.
2. Assemble gaskets, bolts, and joint parts.	2.1 The correct **gaskets**, **bolts**, and **joint** parts are selected based on the specifications. 2.2 The gasket material is checked for compatibility with the fluids or gases that will flow through the system, ensuring proper sealing. 2.3 The bolts are inspected for any defects or damage before use, ensuring they meet the required specifications. 2.4 The gaskets are positioned properly within the joint, ensuring no misalignment or gaps that could lead to leaks. 2.5 Bolts are inserted into the joint and tightened in the correct sequence and to the specified torque to ensure a uniform seal. 2.6 The appropriate tools, such as torque wrenches, are

	used to tighten the bolts to the required torque settings, avoiding over-tightening or under-tightening. 2.7 The assembled joint is checked for any misalignment, ensuring the gasket is compressed evenly and securely.
3. Apply torque and tightening steps.	3.1 The appropriate torque specifications for the bolts, nuts, or fasteners are identified based on the joint design, material. 3.2 The correct tools, such as a torque wrench, are selected and calibrated to ensure accurate torque application. 3.3 The bolts or fasteners are tightened following the correct sequence, ensuring uniform pressure across the joint and preventing misalignment or damage. 3.4 The initial hand tightening of bolts is done to ensure all parts are in place and aligned before applying final torque. 3.5 Torque is applied gradually in multiple steps, starting from the center of the joint and working outward, to ensure even distribution and prevent distortion
4. Install supports, hangers, and guides.	4.1 Types, sizes, and quantities of supports, hangers, and guides are selected based on the project requirements. 4.2 The supports, hangers, and guides are installed according to the specified design and layout. 4.3 Proper spacing between supports, hangers, and guides is maintained according to the design specifications. 4.4 All supports, hangers, and guides are tightened and secured during operation.
5. Verifying alignment and required slope.	5.1 Alignment is verified to ensure the specified dimensions and tolerances as per drawings. 5.2 Tools are used to measure and check the alignment of pipes and components. 5.3 Required slope is confirmed by using appropriate equipment. 5.4 Adjustments are made to the pipe supports, hangers, or guides, as necessary for correct alignment and slope.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Standard Pipes	1.1 ASME B-36.10 (Carbon & alloy steel) 1.2 ASME B-36.19 (Stainless steel)
2. Flanges,	2.1 Threaded 2.2 Socket weld 2.3 Slip-on 2.4 Lab joint 2.5 Weld neck 2.6 Blind
3. Valves,	3.1 Ball

	3.2 Plug 3.3 Glove 3.4 Butterfly 3.5 Needle 3.6 Pressure control
4. Gaskets	4.1 Metallic • Soft iron • Low carbon steel • Stainless steel • Monel • Inconel 4.2 Non-Metallic • Rubber • Teflon • Compressed non-asbestos fiber • PTFE
5. Bolts	5.1 Stud bolds 5.2 Head bolts 5.3 Anchor bolts
6. Joint	6.1 Welded joint 6.2 Threaded joint Use for onshore rigs: 6.3 Regular joint 6.4 Internal flush 6.5 Non-upset 6.6 Full hole 6.7 Hemmer or Weco joint

Curricular Content Guide

1. Underpinning Knowledge	1.1. Understand different types of pipe and Fitting Installation methods 1.2. Types of flanges and valves used in piping systems. 1.3. Different gasket materials and their suitability for specific pipe applications. 1.4. Types and sizes of bolts required for different flanged joints and the correct assembly procedures. 1.5. The importance of applying the correct torque 1.6. Tightening Procedures and Sequence 1.7. Different types of pipe supports, hangers, and guides, and their application. 1.8. Installation procedures for supports, hangers, and guides. 1.9. Importance of correct alignment. 1.10. Tools and methods used to measure and verify the alignment of pipes.
2. Underpinning Skills	2.1. Installing pipes, flanges, valves, and fittings.

	2.2. Ensuring proper alignment of pipes, flanges, and valves. 2.3. Assembling gaskets, bolts, and joint parts. 2.4. Bolting and Tightening Techniques 2.5. Applying the appropriate torque to bolts and fasteners. 2.6. Tightening Procedures and Adjustments 2.7. Installing pipe supports, hangers, and guides. 2.8. Adjusting and positioning supports. 2.9. Checking the alignment of pipes using appropriate tools and techniques. 2.10. Verifying and adjusting the slope of pipes.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Tools, equipment and facilities appropriate to the process or activities. 4.4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed pipes, flanges, valves, and fittings 1.2 assembled gaskets, bolts, and joint parts 1.3 applied torque and tightening steps 1.4 installed supports, hangers, and guides 1.5 verified alignment and required slope.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **UNDERSTAND INSPECTION AND TESTING PROCEDURES**	**Nominal Duration:** 25 Hrs.	**Unit Code:** SICIP-BPI-APF-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand inspection and testing procedures. It specifically includes the tasks of interpreting inspection, performing testing of weld and cleaning and store equipment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Interpret inspection	1.1 Inspection requirements for welding are interpreted. 1.2 Type of inspection of welding are identified. 1.3 Destructive or non-destructive tests of welding are interpreted. 1.4 Inspection report are interpreted from template.
2. Perform testing of weld	2.1 Test sample are selected and collected 2.2 Visual test is carried out according to work place standard. *2.3* Weld job are tested by using ***testing equipment.*** 2.4 ***Destructive*** and ***non-destructive simple tests*** are performed as per job requirement. 2.5 ***Defects*** are identified and recorded.
3. Clean and store equipment	3.1 Equipment and tools are cleaned and stored as per workplace requirements and environmental regulations. 3.2 Wastes are disposed off accordance with workplace requirements. 3.3 Necessary environmental preservation/safeguarding procedure is maintained during waste disposal. 3.4 Wastages are preserved in designated places.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Testing equipment	1. 1 Dye penetration test kit 1. 2 Bending test equipment 1. 3 Fracture test machine 1. 4 Torch light 1. 5 Magnifying glass
2. Destructive test	2. 1 Fracture 2. 2 Bend
3. Non-destructive test	3. 1 Visual 3. 2 Dye-penetrate

4. Defects	4. 1 Lack of penetration 4. 2 Excess penetration 4. 3 Porosity 4. 4 Crack 4. 5 Slag inclusions 4. 6 Spatter 4. 7 Undercut 4. 8 Lack of fusion 4. 9 Irregular shape and dimension

Curricular Content Guide

1. Underpinning Knowledge	1.1. Inspection techniques and the standards. 1.2. Interpret the results of various inspection methods to determine the quality. 1.3. Types of Weld Testing and various testing methods. 1.4. Testing Standards and Criteria. 1.5. Cleaning methods for welding equipment. 1.6. Procedures for storing welding equipment
2. Underpinning Skills	2.1. Performing visual inspections of welded joints. 2.2. Documenting inspection results and generating reports. 2.3. Performing various tests to evaluate the quality and strength of welds. 2.4. Cleaning welding equipment. 2.5. Performing regular maintenance checks on welding machines and tools.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedure 4. 3 Tools, equipment and facilities appropriate to the process or activities. 4. 4 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted inspection

	1.2 performed testing of weld 1.3 cleaned and store equipment.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Advanced Pipe Fitting for Gas and Onshore Rigs**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	AC/DC Arc welding machine	10
5.	Gas cutting set	03
6.	Spare cylinder set (Argon, oxygen and acetylene)	01
7.	Hand grinding machine	10
8.	Bench vice	10
9.	Tool box set for welding workshop	10
10.	Fire extinguisher	04
11.	Welding booth with exhaust system	10
12.	First aid box	03
13.	Baby Grinder (4’’)	04
14.	Chain tong	02
15.	Pipe bender/Tube bender	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Advanced Pipe Fitting for Gas and Onshore Rigs** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.